M-EVENT spec: scheduled economic-event calendar

Rev 2, 2026-10-22 (ET). Rev 1 plus every change in Analyst 2's review (`specs/m-event-review-analyst2.md`, all accepted) and Ettienne's answers to Q1–Q9; see the changelog (§12). VERIFY items resolved for rev 2 cite the URL fetched on 2026-10-22. Checked against `github.com/ettiennehugo/MultiCharts.Net` `main` @ `92a6059` (PR #22 merged; `git pull` was already up to date). Source URLs and standard release times were checked on 2026-10-22. Anything not confirmed from an official page is marked **VERIFY**. Anything about MC behaviour that the Programming Guide does not state is marked **PROBE**.

Sources: the repo (`_Holidays`, `_NyseHolidays`, `_CmeHolidays`, `_CmeSessionClock`, `_RollDetector`, `_DailyContext`, `_TodNormaliser`, `_RegimeEngine`, `_LevelRef`, `_PortMM`/`_PortKeys`, `_CustomSignal`, `_EconomicCalendar.Strategy.CS`, `tests/CalendarTests`), `specs/phase3-specs.md` ("P3"), `specs/missing-functionality-specs.md` ("MFS"), `specs/m-port-spec.md` ("M-PORT"), and `trading-idea-catalogue.md` (Gate 0, windows W-M / W-T, M-NEWS row).

Scope (option 1 only). M-EVENT is a calendar of *when* scheduled events happen. It has no headline sentiment, no consensus, no actual values and no surprise metric. M-NEWS stays deferred. `EAvwapAnchor.NewsBar` gets a calendar-only hook on T1 instants, post-event only (EV-6.12); full wiring is rev 1.1.

Not touched. `_EconomicCalendar.Strategy.CS` (the old FX CSV trader with `[Input] EventFile`) stays as it is. M-EVENT is a base service, not a strategy (Q7, decided).

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0. Cross-cutting rules (inherited)

- **EV-0.1 Laziness (MFS §0.1, P3 §0.1).** The engine is an `internal sealed class` created in `StartCalc()` only when `EventConfig.enabled` is true, and is `null` otherwise. **No new `[Input]`s** on `_CustomSignal` or any of the 14 strategies on it. A disabled engine's accessors return neutral values (`false`, `NaN`, `EEventTier.None`, empty lists) and log one Warning (the `m_regimeQueryWarned` pattern).
- **EV-0.2 MC-free core.** All date rules, tables and lookups live in `_EventCalendar.Function.CS` inside `// <event-rules>` … `// </event-rules>` and `// <event-table>` … `// </event-table>`. They use no MultiCharts types, so `tests/EventTests/gen.sh` can extract and compile them (the `tests/CalendarTests` pattern).
- **EV-0.3 C# 7.3.** No switch expressions, `??=`, nullable reference types or ranges. No LINQ and no allocation on the per-bar path: lookups are binary searches over sorted arrays built once per year.
- **EV-0.4 Journal.** Facts go to `engineSink(ETraceLevel.Journal)` and are stamped with bar times (`yyyy-MM-dd HH:mm:ss`, chart TZ), never bar indexes. Per-bar detail goes to `Debug`. A `null` sink formats no string. Event instants in journal lines are printed in ET with the zone label (`08:30 ET`) next to the chart-TZ bar stamp.
- **EV-0.5 Built-ins first.** Time zones come from `TimeZoneInfo` via the existing `findTz`, `BarTimeZone`, `ToZone` and `resolveBarTimeZone`. Holidays and half days come from `_NyseHolidays.Rules` and `_CmeHolidays.Rules` (MC's Quote Manager holiday lists are not readable from code, so nothing new is invented here). Rolls come from `rollOn(1)` / `_RollDetector`. Trading days come from `tradingDayKey`.

- **EV-0.6 Config hash.** `EventConfig.Canon()` returns its fields in fixed order, plus a hash of the loaded table and CSV (EV-4.6), so the journal's per-run config hash identifies the calendar version (P3 §0.5).
- **EV-0.7 Recalc idempotence.** Every query is a pure function of (bar time, config, table). There is no state that a recalc could double-count.

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1. Events and tiers

1.1 Macro events (official schedules)

#	Kind (`EEventKind`)	Agency, schedule source	Standard time (ET)	Default tier	Generated by
EV-1.1	`FomcDecision` (flag Sep = SEP / dot plot)	Fed Board, FOMC calendar (2021+ and the next year); historical materials by year	14:00 statement, on the last day of the meeting	T1	table. The Fed page marks tentative meetings and notation votes; only meetings with a published statement are `FomcDecision`. Pre-2013 statements were at 14:15 ET, and at 12:30 ET on 2011–2012 SEP days (**VERIFY**), which matters only for a coverage extension (EV-4.6)
EV-1.2	`FomcPress` (Chair press conference)	same page ("Press Conference" link)	14:30	T1	table (every meeting since Jan 2019; 2011-04 → 2018 only the quarterly SEP meetings; read per year from the archive). Rule resolved: Fed FEDS Note 2021-10-12 ("Between 2011 and 2018 ... only after FOMC meetings associated with a SEP ... starting in January 2019 ... after every FOMC meeting"); first all-meeting presser 2019-01-30 14:30 ET per Fed calendar Jan 2019
EV-1.3	`FomcMinutes`	same page ("Released ")	14:00	T2	table (three weeks after the decision, per the Fed page)
EV-1.4	`Cpi`	BLS, release schedule / archive `bls.gov/`	08:30	T1	table

#	Kind (`EEventKind`)	Agency, schedule source	Standard time (ET)	Default tier	Generated by
		`schedule/<yyyy>/`			
EV-1.5	`Ppi`	BLS	08:30	T2	table
EV-1.6	`Nfp` (Employment Situation)	BLS	08:30	T1	table. NFP is released on Good Friday when it falls on one, while NYSE is closed and CME equity trades an abbreviated session. Resolved: 2021-04-02 and 2026-04-03, both 08:30 ET
EV-1.7	`Jolts`	BLS	10:00	T3	table
EV-1.8	`Claims` (UI weekly claims)	DOL ETA, claims page ("each week on Thursday … 8:30am", with listed holiday exceptions, e.g. Wed 2026-11-25)	08:30 Thursday	T3	dates taken from the DOL release archive per week; the Thursday rule is only the fallback for future weeks
EV-1.9	`Pce` (Personal Income and Outlays)	BEA, release schedule	08:30	T2	table
EV-1.10	`GdpAdvance`, `GdpSecond`, `GdpThird`	BEA	08:30	T2 / T3 / T3	table
EV-1.11	`RetailSales` (Advance Monthly Retail and Food Services)	Census, indicator calendar, retail schedule	08:30	T2	table
EV-1.12	`IsmMfg`	ISM, release calendar ("first business day … 10:00 a.m."; ISM-listed exceptions, e.g. Jan 2026 → Mon 01-05)	10:00	T2	rule, checked against the published table. Historical ISM dates before the current calendar come from Wayback captures of the ISM calendar. Where no capture exists, the rule output is flagged `RuleOnly` and reported in the coverage line
EV-1.13	`IsmSvc`	ISM ("third business day … 10:00 a.m.")	10:00	T3	rule, checked against the published table. Historical ISM dates before the current calendar come from Wayback captures of

#	Kind (EEventKind)	Agency, schedule source	Standard time (ET)	Default tier	Generated by
					the ISM calendar. Where no capture exists, the rule output is flagged `RuleOnly` and reported in the coverage line
EV-1.17	`JacksonHole` (Chair speech, KC Fed symposium)	KC Fed / Fed Board speech calendar	per announcement	T2	table; `Scheduled` only from its first announcement (EV-4.4(f)); **VERIFY** source page per year at build
EV-1.18	`ChairTestimony` (semiannual Monetary Policy Report)	Fed Board calendar, congressional committee notices	per announcement	T2	table; `Scheduled` only from its first announcement (EV-4.4(f))
EV-1.19	`EiaPetroleum` (Weekly Petroleum Status Report, for CL)	EIA, WPSR schedule ("10:30 a.m. eastern time on Wednesday with the following exceptions"; holiday weeks move, e.g. Thu 2026-11-12 12:00 for Veterans Day, Mon 2025-12-29 17:00 for Christmas)	10:30 Wednesday	T2 (CL); class `Energy`	rule + EIA exception table; historical exceptions from Wayback captures of the schedule page (**VERIFY** at build; the live page lists only 2025-01 → 2026-11). Holiday-week retimes follow the EV-5.1 known-from rule: known from the first schedule-page capture showing the move, never `MinValue`

- **EV-1.14 Coincident releases.** CPI with Real Earnings, GDP with PCE (on BEA's combined days), NFP with Claims (when a Thursday NFP happens) are separate events at the same instant. `EventsOn` returns both, and window and tier queries use the **max** tier at that instant.
- **EV-1.15 Tier override.** `EventConfig.tierOverride` (a `Dictionary<EEventKind, EEventTier>`) replaces the default tier. It is part of `Canon()`. The defaults above are accepted (Q1) as a prior, not a finding; `tierOverride` stays in `Canon()`.
- **EV-1.16 Times are ET wall-clock.** Every macro event is stored as (ET date, ET time of day). It is never stored as an offset or in UTC (see §3).

1.2 ES calendar items (rule-generated)

#	Kind	Rule	Instant (ET)	Tier
EV-1.20	`QuarterlyExpiry`	Third Friday of Mar/Jun/Sep/Dec. If that day is not an NYSE trading day, the previous NYSE trading day (e.g. ESM26 Thu 2026-06-18, because Juneteenth is Fri 06-19; ESM27 Thu 2027-06-17; ESH08 Thu 2008-03-20). Authority: CME per-contract ProductCalendar and `tools/`	09:30	C

#	Kind	Rule	Instant (ET)	Tier
		`m-event/nyse_holidays.txt`; no override table. Final settlement is the Special Opening Quotation. **Normative**		
EV-1.21	`RollWeek` (span)	CME customary roll date = "the Monday prior to the third Friday of the expiration month" (CME roll dates) through expiry. CME's page (checked 2026-10-22) states the Monday rule for 2025–2028. Earlier years use the rule then in force, stored **by date range**: "the Thursday eight calendar days before expiration" (resolved from a CME notice for 2011, nexusfi copy, and the page wording quoted in ohlcv.io). The cutover (the secondary source says the page wording changed between 2022 and 2023) is **VERIFY** via Wayback captures of `rolldates.html`. The actual dxFeed @ES splice comes from `rollOn(1)`, not from this rule (EV-5.4)	span: roll-date 00:00 → expiry 09:30	C
EV-1.22	`MonthlyOpex`	Third Friday of every month; NYSE holiday → preceding business day (Good Friday cases). In quarterly months it coincides with EV-1.20 (flag `Quad`). Monthly SPX AM-settled at 09:30; SPXW PM at 16:00	09:30 (monthly SPX AM settlement, every month) and 16:00 (SPXW PM); `Quad` only flags the quarterly coincidence	C
EV-1.23	`IndexRebalance`	S&P DJI quarterly rebalance "after the close on the third Friday of March, June, September, and December" (S&P U.S. Indices methodology); effective before the next open. Off-cycle constituent changes are **not** scheduled and go only through the CSV (EV-4.4)	16:00 on the third Friday	C
EV-1.24	`HalfDay`	`_CmeHolidays.Rules.Get(d)` with `RthCloseEt != null` (early close 13:15 ET) or `NoRth && HaltEt != null` (holiday halt). It is a day classification, not a new list: the calendar is the existing one	`RthCloseEt` or `HaltEt`	C
EV-1.26	`VixExpiry`	Cboe VIX futures/options final settlement: Wednesday 30 days before the next month's standard SPX expiry, moved for holidays (**VERIFY** the Cboe rule text and holiday handling at build)	09:30 (SOQ)	C

#	Kind	Rule	Instant (ET)	Tier
EV-1.27	`RussellRecon` (optional, enabled off by default)	FTSE Russell annual reconstitution, effective after the close on the last Friday of June (**VERIFY** per year on FTSE Russell's schedule, which has varied)	16:00	C

- **EV-1.24a Class mask.** `[Flags] enum EEventClassMask : byte { None = 0, Macro = 1, Calendar = 2, Energy = 4, All = Macro | Calendar | Energy }`. `EiaPetroleum` is class `Energy`. ES macro windows (`IsInWindow`, `EventBlackout`, the `EventConfig.classes` default) default to `Macro` / `Macro | Calendar` and exclude `Energy`; A2-20 selects `Energy` explicitly.
- **EV-1.25 Tier enum.** `enum EEventTier { None = 0, T3 = 1, T2 = 2, T1 = 3, C = 8 }`. Comparisons such as "tier ≥ T2" apply to macro tiers only. C (calendar) is a separate class selected by `EEventClassMask`, so a no-trade window around T1 never pulls in OpEx by accident.

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2. Data model

- **EV-2.1 Record** (in `// <event-rules>`):

```
public struct EventRec {
    public EEventKind kind; public EEventTier tier; public EEventFlags flags;  // Sep, Qu
    public DateTime etTime;        // ET wall clock (Kind=Unspecified); for spans, the st
    public DateTime etEnd;         // spans only (RollWeek); else == etTime
    public DateTime utc;           // derived once, EV-3.2
    public DateTime knownSinceUtc; // when the schedule for this instant was public (EV-5
    public DateTime canceledSinceUtc; // MaxValue = never canceled (EV-5.3)
    public DateTime tradingDay;    // Globex trade date of the instant (EV-3.4)
    public string refPeriod;       // "2025-09" / "2026Q3"; "" for calendar items
    public string source;          // short source tag, e.g. "BLS-2025-lapse", "FED-20200
}
```

- **EV-2.2 Year cache.** `_EventCalendar.Year(int y)` builds a sorted `EventRec[]` for year y once: table + rules + corrections + CSV (EV-4). The engine keeps years y−1, y, y+1 (the `_Holidays` "current and previous year" rule, extended forwards for `NextEvent` across New Year).
- **EV-2.3 Sorting.** The cache is sorted by `utc`, then by tier descending. The day index is a `Dictionary<DateTime, int[]>` from trading day to record indexes, built with the year.

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3. Time zones and DST

- **EV-3.1 Store ET wall time.** "08:30" means 08:30 America/New_York on that date, whether EST or EDT. Windows id `"Eastern Standard Time"` (the id `_CmeSessionClock` and `autoTradingDay` already use) goes through `findTz`.
- **EV-3.2 Derived UTC.** `utc = TimeZoneInfo.ConvertTimeToUtc(etTime, est)`. Release times are never inside the 02:00–03:00 transition hour. A record that is would be a table error: `Configure()` rejects it if `est.IsInvalidTime` or `IsAmbiguousTime` (no clamping).
- **EV-3.3 Bar comparison.** A bar's interval in UTC is `[barOpenUtc, barCloseUtc)`. `barCloseUtc` comes from `Bars.Time[0]` (MC stamps bars at their close) converted from `BarTimeZone` with the same `IsInvalidTime` guard as `ToZone`. `barOpenUtc = barCloseUtc − Resolution.Duration()` for time-based bars. For non-time bars it is the previous bar's close (**PROBE** the first bar of a session). The event "is in" the bar whose interval

contains `utc`. So a 1m bar stamped 08:31 ET contains the 08:30:00 release, and the bar stamped 08:30 is the last pre-event bar. PROBE P-EV1: verify empirically that the 1m volume spike of 08:30 releases lands in the bar stamped 08:31 ET for dxFeed @ES in all three `RequestTimeZone` settings, including DST and mismatch weeks (test EV-8.11).
- **EV-3.4 Minutes are UTC differences.** `MinutesTo` and `MinutesSince` are `(utc − nowUtc).TotalMinutes`, never ET wall-clock subtraction. Across the DST weekend, Fri 2026-03-06 14:00 ET to Mon 03-09 08:30 ET is 3 × 1440 − 330 − 60 = 3930 minutes, not 3990 (test EV-8.3).
- **EV-3.5 Chart zones.** All three `RequestTimeZone` cases work through `resolveBarTimeZone`: Exchange (CME = Central), Local (the user's PC, which may observe no US DST, so the ET offset to local time changes on the US DST dates and the window must follow it), and GMT. US/EU DST mismatch weeks (March and late Oct–early Nov) are where a fixed offset would break. Tests EV-8.3 cover them. On a pre-2007 extension, Windows `TimeZoneInfo` historical adjustment rules must be confirmed (US DST rules changed in 2007).
- **EV-3.6 Trading day.** `tradingDay` = the Globex trade date of the first Globex session open at or after the instant, by `_CmeSessionClock` (an instant ≥ 18:00 ET on a session day maps to the next date; an instant in the 17:00–18:00 break, on a weekend or on a closed day maps to the date of the next session that opens). Sun 2020-03-15 17:00 ET → Mon 2020-03-16 (EV-8.5).
- **EV-3.7 Daily and weekly bars.** On a daily or weekly Data1 (or `DailyConfig` stream), instants mean nothing inside the bar. Only the day queries (EV-6.4/6.5) are valid. The window queries return `false` and log one Warning if `Resolution.Duration() ≥ 1 day`. A weekly bar's classification is the max over its trading days (`EventsInRange`).

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4. Building the list (mirrors `_NyseHolidays` / `_CmeHolidays`)

The repo's pattern: date **rules** and a short **special-dates array** with a source comment per date, in a nested MC-free `Rules` class, generated per year, wrapped by a `_Holidays`-style host and tested in `tests/CalendarTests`. M-EVENT follows it.

- **EV-4.1 Rule-generated kinds** (`Rules.Generate(year)`): `Claims` (dates from the DOL archive (EV-1.8); the rule only generates beyond the archive), `IsmMfg`/`IsmSvc` (Nth business day with NYSE holidays from `_NyseHolidays.Rules`, plus `s_ismExceptions` (e.g. Mfg Jan 2026 → 01-05) for ISM-published moves, e.g. Services Jul 2026 → Mon 07-06), every §1.2 item, `EiaPetroleum` (Wednesday rule + `s_eiaExceptions` from the EIA schedule), and `HalfDay` from `_CmeHolidays.Rules`.
- **EV-4.2 Table kinds** (FOMC ×3, CPI, PPI, NFP, JOLTS, PCE, GDP ×3, Retail, Jackson Hole, Chair testimony): these have no reliable rule, so they go in a **generated in-code table**, `// <event-table>`: `static readonly long[] s_yyyymmddhhmm` plus a parallel `byte[] s_kind`, one pair per year (an `int` cannot hold 202610220830 > 2^{31}), with a header comment giving the build script version, the fetch date and per-source SHA-256 of the pages fetched. The table is written by the script in EV-4.5, never by hand, and the repo diff is what Analyst 2 reviews.
- **EV-4.3 Corrections list** (`s_corrections`, in code, hand-maintained, every row with a source URL): past-dated changes to the scheduled calendar. Each row has (kind, refPeriod, originalEt or none, actualEt or Canceled, announcedEt, flag). See §5.
- **EV-4.4 CSV override / extension** (`EventConfig.csvPath`, empty = none). Columns: `action,kind,date_et,time_et,tier,flags,ref_period,announced_et,source`. `action` ∈ {`add`, `remove`, `retime`, `retier`}. `kind` may also be `Custom:<name>` (a free-form kind, tier required). Rules: (a) CSV rows apply after table, rules and corrections; (b) the file is read once in `Configure()` with no per-bar I/O; (c) any parse error fails `Configure()` with line number and text (report, never skip); (d) every applied row is flagged `Csv` and counted in the journal; (e) the file's SHA-256 goes into `Canon()`. A `retime` row with `announced_et` later than its original date is treated as an unscheduled change (EV-5.2). (f) `announced_et` is

required for add and retime rows. A blank value is allowed only with flag Scheduled plus a source showing publication before the instant; otherwise Configure() fails.

- **EV-4.5 Build script** tools/m-event/build_table.py (outside MC; Python 3, stdlib + requests/lxml). Steps: (1) fetch the official pages per year (Fed fomccalendars.htm and fomchistoricalYYYY.htm via fomc_historical_year.htm; BLS bls.gov/schedule/YYYY/ and the monthly MM_sched.htm pages; BEA bea.gov/news/schedule plus the BEA release archives per release; Census calendar-listview.html per year and retail/release_schedule.html); (2) parse to (kind, ET date, ET time, refPeriod, sourceURL); (3) **cross-check** each date with a second independent source, St. Louis Fed ALFRED/FRED release calendars (fred.stlouisfed.org/releases/calendar?rid=<id>&y=<yyyy>, e.g. rid 9 = Advance Retail Sales) and, for FOMC, the statement press-release URL (newsevents/pressreleases/monetaryYYYYMMDDa.htm); (3b) **first-publication check:** for each year and source, fetch the earliest Wayback capture of the annual schedule published before the year starts, plus captures through the year. Any instant that differs from today's page without a corrections row is reported as a silent reschedule. Its announcedEt is the first capture showing the new date, an upper bound and therefore conservative; (4) write the // <event-table> region and tools/m-event/out/events_<from>_<to>.csv (a human-readable copy); (5) print a disagreement report and refuse to write if any (kind, refPeriod) has two dates without a matching corrections row. (6) Store the raw fetched pages (or capture URLs with timestamps) under tools/m-event/raw/ so the build is reproducible (page SHAs alone cannot be rebuilt once the pages change). (7) Resolve every remaining **VERIFY** in this spec (list in §12.2) and print each with its source. **No date is typed by hand or taken from a commercial calendar.**
- **EV-4.6 Coverage.** At least **2016-01-01 → 2027-12-31**, which covers the 1m window W-M (Jun/Jul 2016 →) and the tick window W-T (Jul 2024 →) with margin, plus the 2027 schedules the Fed and the agencies have already published (the Fed page lists 2027 meetings; BLS and Census publish next year's schedule late in the year: **VERIFY** the 2027 BLS/BEA availability at build time). Configure() journals EVENT coverage(<kind>: first..last) and returns an error if the chart's first bar is before the coverage start of an enabled kind (EventConfig.allowPartialCoverage = false by default). Coverage decided (Q2): 2016–2027 now, designed to extend back to 2000. An extension requires EV-5.6 (the Oct 2013 lapse, the 2001/2008 intermeeting FOMC actions), the pre-2013 14:15 ET FOMC statement time (EV-1.1) and the pre-2007 TZ check (EV-3.5).
- **EV-4.7 Year rollover live.** Outside coverage (e.g. 2028 before the table update) table kinds are **absent**, never guessed. Each year the engine is asked for, a Warning EVENT <year> table kinds missing: <list> is logged once. Rule kinds still generate.
- **EV-4.8 Table update procedure.** Re-run the script (late each year, and after any agency notice), commit, run tests/EventTests. The script's report goes in the PR body.

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5. Backtest honesty: known-in-advance only

- **EV-5.1 knownSinceUtc.** A pre-event window or MinutesTo value at time *t* may use a record only if t ≥ knownSinceUtc. Annual schedules (Fed, BLS, BEA, Census, ISM, EIA) are published months ahead, so their rows get MinValue. This holds only for instants that match the schedule's first-publication capture (EV-4.5 step 3b). Instants that moved after first publication get knownSinceUtc = the first capture showing the move. A year with no capture before Jan 1 gets knownSinceUtc = its earliest capture, journaled as EVENT knownSince capture-bound <source> <year>.
- **EV-5.2 Unscheduled = known only at the instant.** A record flagged Unscheduled has knownSinceUtc = announcedEt (or utc when there was no notice). It is **excluded** from NextEvent, MinutesTo and pre-event windows before then. It is **included** in EventsOn,

`PrevEvent`, `MinutesSince` and post-event windows from its instant (it happened; a post-event no-trade rule may honestly react to it).

- **EV-5.3 Rescheduled and canceled.** For a lapse in appropriations, every release scheduled inside the lapse gets `canceledSinceUtc` = the lapse start (00:01 ET of the first unfunded day; agency contingency plans are public), and its replacement instant gets `knownSinceUtc` = the earliest verifiable publication of the revised date (upper bound). For other reschedules, `canceledSinceUtc` = `knownSinceUtc` of the new instant = the announcement. A `Canceled` row with no replacement (e.g. Oct 2025 CPI) behaves the same way with no new instant. Before `canceledSinceUtc` the original instant is a normal known-in-advance record. Unscheduled FOMC is known at its instant (EV-5.2).
- **EV-5.4 Roll.** `RollWeek` is a schedule (known in advance). The **actual** @ES splice day is data and comes only from `rollOn(1)`. `IsRollDay` reports the splice as it is detected (same bar, no lookahead). `IsRollWeek` reports the CME rule. They can differ. Q3 (decided): do not wait for live; the build plan tabulates `rollOn(1)` splice dates over 2016–2024 history against the date-ranged CME rule (EV-1.21) and journals the weekday distribution (§10 step 7).
- **EV-5.5 Known items to put in `s_corrections`** (each verified against the cited page before commit; the script's disagreement check enforces this):

#	Event	Treatment	Source
EV-5.5a	FOMC unscheduled meeting, statement Tue **2020-03-03 10:00 ET** (Fed press release)	`FomcDecision`, `Unscheduled`, known at the instant	`federalreserve.gov/newsevents/pressreleases/monetary20200303a.htm`; listed as "March 2 (unscheduled) Meeting ... Statement (Released March 3, 2020)" on fomchistorical2020.htm
EV-5.5b	FOMC unscheduled meeting Sun **2020-03-15**, statement released 17:00 ET (resolved: "For release at 5:00 p.m. EDT", monetary20200315a.htm; trade date Mon 03-16 by EV-3.6)	`Unscheduled`	`monetary20200315a.htm`, `fomcminutes20200315.htm`
EV-5.5c	Regular FOMC of 2020-03-17/18 not held as scheduled (superseded by 03-15)	`Canceled`, canceledSince = 2020-03-15 17:00 ET. Cancellation resolved: "March 17-18 (cancelled) Meeting - 2020" on fomchistorical2020.htm. The page gives no cancellation notice time; 03-15 17:00 is the design choice (**VERIFY** whether an earlier notice exists)	`fomchistorical2020.htm`
EV-5.5d	2025 lapse in appropriations (BLS): e.g. Sep 2025 Employment Situation, Fri 2025-10-03 → Thu **2025-11-20** 08:30; Oct 2025 Employment Situation and Oct 2025 CPI	`Delayed/Canceled`; EV-5.3 lapse rule: `canceledSinceUtc` = 2025-10-01 00:01 ET (lapse 2025-10-01 → 11-12, per the BLS CPI shutdown	BLS revised release dates

#	Event	Treatment	Source
	canceled; Nov 2025 NFP 12-05 → **12-16**; Nov 2025 CPI 12-10 → **12-18**; Sep 2025 PPI 10-16 → **11-25**; Sep 2025 CPI 10-15 → **Fri 10-24 08:30** (BLS recalled staff; resolved: BLS notice, last modified 2025-10-10, and the release; this row is **not** on the revised-dates page, so `knownSinceUtc` upper bound = 2025-10-10 per the notice's last-modified date, **VERIFY** the exact time from a Wayback capture); Sep 2025 JOLTS canceled; Oct 2025 JOLTS 12-02 → **12-09**; Oct 2025 PPI canceled; Nov 2025 PPI 12-11 → **2026-01-14**; Dec 2025 PPI 01-14 → **01-30**	FAQ); `knownSinceUtc` of each new date = its first Wayback capture on the revised-dates page (the page has no revision history: **VERIFY** per row at build). All dates in this row were re-checked against the page on 2026-10-22 and agree	
EV-5.5e	2026 lapse (BLS): JOLTS Dec 2025 02-03 → 02-05; Employment Situation Jan 2026 Fri 02-06 → **Wed 02-11**; CPI Jan 2026 02-11 → **Fri 02-13**; PPI Jan 2026 02-12 → **02-27**; PPI Feb 2026 03-12 → **03-18** (all five rows confirmed on the page 2026-10-22; the PPI rows are listed in the page's 2025 table)	as above; the 2026 lapse start date and the per-row announcement dates are **VERIFY**	same page
EV-5.5f	2025 lapse effects at BEA and Census (GDP Q3 2025, PCE, retail sales Sep–Nov 2025)	to be compiled by the script from BEA/Census notices; not listed here because they were not verified for this spec (**VERIFY**)	bea.gov, census.gov notices
EV-5.5g	Dec 2018 – Jan 2019 partial shutdown (Commerce: BEA GDP Q4 2018, Census retail sales; BLS was funded). Resolved: Q4-2018 GDP advance (orig. 2019-01-30) and second (orig. 02-28) **replaced by one "initial" estimate Thu 2019-02-28 08:30** (BEA release), announced 2019-02-06 (BEA blog); Dec 2018 retail sales 2019-01-16 → **Thu 2019-02-14 08:30** (Census revised	GDP advance → `Canceled`, second → retimed `GdpAdvance`-tier record flagged `Delayed`; EV-5.3 lapse rule	BEA/Census archives

#	Event	Treatment	Source
	calendar, first issued 2019-01-30 per its note, revised 02-28). Lapse start/end dates, the PCE rows and the Census tip-sheet first-issue content are **VERIFY**		
EV-5.5h	Days of mourning 2018-12-05, 2025-01-09 (already in `_NyseHolidays`/ `_CmeHolidays`): any release that moved because of them (e.g. claims)	script check against DOL/ agency pages	—

- **EV-5.6 Out of range.** Out of range unless coverage is extended (Q2). If it is, add the Oct 2013 lapse, the 2001-01-03, 2001-04-18, 2001-09-17, 2008-01-22 and 2008-10-08 intermeeting FOMC actions (**VERIFY**), and the pre-2013 FOMC times (EV-1.1).
- **EV-5.7 Audit journal.** On `Configure()`, one line per correction applied inside the chart's range: `EVENT correction <kind> <ref> <orig>-><actual|canceled> known(<announced>) src(<tag>)`.

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6. API

Classes and files. `_EventCalendar.Function.CS`: `EventConfig`, `EEventKind`, `EEventTier`, `EEventFlags`, `EEventClassMask`, `EventRec`, `EventDay`, the static `Rules` (regions), and the `internal sealed class _EventCalendar`. Host code goes in `_CustomSignal`, next to M-DAILY / G-R.

- **EV-6.1 Config** (protected, no `[Input]`; set in a derived ctor or `Create()`):

```
public sealed class EventConfig {
    public bool enabled = false;
    public EEventClassMask classes = EEventClassMask.Macro | EEventClassMask.Calendar;
    public EEventTier minTier = EEventTier.T3;          // records below are dropped at b
    public Dictionary<EEventKind, EEventTier> tierOverride = new Dictionary<EEventKind, E
    public string csvPath = "";                          // EV-4.4
    public bool allowPartialCoverage = false;            // EV-4.6
    public int blackoutPreMin = 0, blackoutPostMin = 0;  // EV-6.7; 0/0 = no blackout
    public EEventTier blackoutTier = EEventTier.T1;
    public bool blackoutFlatten = false;                 // EV-6.7
    public bool publish = false;                         // EV-7
    public string Validate(); public string Canon();
}
```

- **EV-6.2 Host members on `_CustomSignal`:** `protected EventConfig EventConfig { get; } = new EventConfig();`, `private _EventCalendar m_event;`, `private bool m_eventQueryWarned;`, `internal _EventCalendar EventCalendar { get; }` (Warning once when null), `startEvent()` in `StartCalc` after `startDaily()`/ `startRegime()` (`journal = engineSink(ETraceLevel.Journal)`, `warnOnce = engineSink(ETraceLevel.Warning)`, `Configure()` errors throw `ArgumentException`), and `stepEvent()` in the engine step after `stepRegime()` (it only feeds the bar UTC interval and the roll flag; closed bars only, idempotent per `CurrentBar`).

- **EV-6.3 Engine time input.** `stepEvent` computes `nowUtc` (bar close) and `barOpenUtc` once per bar (EV-3.3) and passes them in. The engine never reads a clock (`DateTime.Now` is forbidden, so live and backtest behave the same).
- **EV-6.4 `IList<EventRec> EventsOn(DateTime tradingDay, EEventClassMask mask = All).`** Returns **as-of** records for that Globex trade date (`knownSinceUtc` ≤ current bar close; Canceled excluded once `canceledSinceUtc` ≤ the bar close), sorted by time. `EventsOnExPost(day, mask, includeCanceled)` is the research-only variant. `EventsInRange(fromDay, toDay)` (as-of) and `EventsInRangeExPost` do the same for ranges (weekly bars).
- **EV-6.5 `EventDay Classify(DateTime tradingDay).`** A struct: `maxTier` (macro), `kindsMask` (ulong), `hasFomc`, `hasSep`, `isHalfDay`, `isRollWeek`, `isQuarterlyExpiry`, `isOpex`, `isRebalance`, `firstReleaseEt` (TimeSpan?). Used as a regime or filter. **As-of rule:** only records with `knownSinceUtc ≤ asOfUtc` count (default `asOfUtc` = the session's 18:00 ET open), so an emergency FOMC never makes its own day an "event day" ex ante (EV-5.2). `ClassifyExPost(day)` includes them and is for research labelling only (§8).
- **EV-6.6 Next / previous.** `bool NextEvent(DateTime nowUtc, EEventTier minTier, EEventClassMask mask, out EventRec e)` (strictly after `nowUtc`, honouring `knownSinceUtc` and Canceled), `bool PrevEvent(...)` (at or before `nowUtc`, including Unscheduled), `double MinutesTo(minTier, mask)` and `double MinutesSince(minTier, mask)` for the current bar (`NaN` when there is none in the cache horizon; the horizon is y+1). Spans (`RollWeek`) are reported by `InSpan(kind)`, not by Next/Prev.
- **EV-6.7 `bool IsInWindow(EEventTier minTier, int preMin, int postMin, EEventClassMask mask = Macro).`** True when the current bar interval `[barOpenUtc, barCloseUtc)` overlaps the half-open `[utc − pre, utc + post)` of any qualifying event (pre-side only for known-in-advance records, post-side for all, EV-5.2). Convenience: `protected bool EventBlackout` = `IsInWindow(blackoutTier, blackoutPreMin, blackoutPostMin)`. **The base does not block orders.** Consumers decide. `blackoutFlatten` only exposes `EventFlattenNow` (true on the first bar entering a blackout) for a consumer's exit code. Wiring it into `_CustomSignal`'s order path is out of scope (Q5, decided: no entry blocking in rev 2).
- **EV-6.8 Roll helpers.** `bool IsRollWeek` (EV-1.21 rule), `bool IsRollDay` (the actual `rollOn(1)` splice seen today), `int SessionsToExpiry` (Globex sessions until the next quarterly expiry, from `_CmeHolidays`).
- **EV-6.9 Neutral values when disabled:** `EventsOn` empty, `Classify` default (`maxTier = None`), `MinutesTo/Since = NaN`, `IsInWindow = false`, `EventBlackout = false`, plus one Warning.
- **EV-6.10 Journal lines** (Journal level, bar-time stamped): on configure, `EVENT created classes(..) minTier(..) csv(<path> rows n sha ..) coverage(..)`; once per trading day with events, `EVENT day <yyyy-MM-dd> tier(<max>) kinds(<list>) firstRelease(<ET>)`; on blackout enter and leave, `EVENT blackout on|off <kind> <ET time> pre(..) post(..)`. when `EventConfig.enabled`, `CS-1` trade-journal rows get two extra columns, `evtDayTier` and `evtMinSince` (NaN-safe; schema unchanged for runs with M-EVENT off; the RUN line records the schema), so Gate 3 event studies can be split later without rerunning (Q8).
- **EV-6.11 Performance.** Classify and window queries are O(log n) with no allocation. `EventsOn` returns a cached read-only list per day.
- **EV-6.12 NewsBar AVWAP anchor hook (Q6).** `bool LastT1Instant(DateTime nowUtc, out DateTime etInstant)` returns the most recent T1 macro instant at or before `nowUtc` (post-event only, as `PrevEvent`; never a future instant). `EAvwapAnchor.NewsBar` will anchor at the bar containing that instant (EV-3.3). Calendar-only: no headlines. Rev 2 specifies and tests the hook (EV-8.12); wiring it into the AVWAP engine is rev 1.1.

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7. Portfolio Trader publishing (M-PORT pattern)

- **EV-7.1 Gate.** Only when `PortfolioPublish && EventConfig.publish && m_event != null`. Chart runs publish nothing (M-PORT §0).
- **EV-7.2 Keys,** added to `_PortKeys` as optional keys (like RegimeP; `Version` stays 1, and the MM must treat missing keys as unknown):

Key	Type	Written	Meaning
`pm.evtTier`	int	first bar of each trading day (idempotent)	`Classify(today).maxTier` as-of (EV-6.5); 0 = none/unknown
`pm.evtMask`	long	same	`kindsMask` cast from `ulong`; ≤ 63 kinds asserted in `Validate()`
`pm.evtWin`	int	every closed bar	1 if `EventBlackout`, else 0
`pm.evtNext`	double	every closed bar	`MinutesTo(blackoutTier)`; NaN = unknown

- **EV-7.3 Book-level use is out of scope.** `_PortfolioRiskManager` does not act on these keys in this spec. Using them in the MM (for example, cutting heat on T1 days) is a separate M-PORT change (Q5, decided: none in rev 2). Because every ES strategy computes the same calendar, the MM may read any one of them. A single shared calendar in `CommonPortfolioData` is **not** done, because write order across strategies is not guaranteed (M-PORT §4).

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8. Tests (`tests/EventTests`, CalendarTests style)

`gen.sh` extracts `// <event-rules>` and `// <event-table>` from `_EventCalendar.Function.CS`, plus `// <calendar-rules nyse>` and `// <calendar-rules cme>` (dependencies), into `Generated/`. `Program.cs` uses `Check(bool, string)` and pass/fail counts. `run.sh` runs `dotnet run -c Release` with `LangVersion 7.3`. Expected dates in the tests are typed from the cited official pages, with the URL in a comment beside each block.

- **EV-8.1 Rules.** ISM: 2026 Mfg/Svc dates equal the ISM 2026 table (Jan 5/7 … Dec 1/3, Svc Jul 6). Claims: every Thursday, with Wed 2026-11-25 replacing Thanksgiving. OpEx/expiry: 2026-06-18 (Thu, Juneteenth), Good Friday on a third Friday: 2022-04-15 → Thu 2022-04-14 (assert; 2026-04-03 is a control that is not a third Friday). Roll dates equal the CME table for 2025–2028. Good Friday NFP (2021-04-02) is generated and kept. EIA: Wed 10:30 by rule, with 2025–2026 exceptions equal to the EIA schedule page.
- **EV-8.2 Table integrity.** Over coverage: 8 scheduled FOMC decisions per year (plus Unscheduled extras); SEP flags on 4 per year; minutes = decision + 21 days (Fed page wording), with a **reported** mismatch rather than a failed test where the Fed's actual dates differ (e.g. holidays); one CPI/PPI/NFP per refPeriod unless a correction cancels it; no release on a **federal** holiday except via a correction; releases on NYSE-only closures (Good Friday) are allowed and listed in the report; every time is in {08:30, 10:00, 10:30, 14:00, 14:30} unless flagged; no duplicates.
- **EV-8.3 DST.** (a) An 08:30 ET event on the Monday after spring-forward (2026-03-09) and fall-back (2026-11-02) maps to 13:30 / 12:30 UTC after the change, and the minute before it is pre-event, for chart zones Eastern, Central (Exchange), UTC, and a no-DST local zone (South Africa Standard Time) and an EU zone (W. Europe) during the US/EU mismatch weeks (e.g. 2026-03-09 … 03-27 and 2026-10-26 … 10-30). (b) `MinutesTo` across the DST weekend equals the UTC difference (EV-3.4 example). (c) A table row at 02:30 ET on a spring-forward Sunday fails `Configure()`.

- **EV-8.4 Bar interval.** On 1m: the bar stamped 08:31 contains an 08:30 release, and the bar stamped 08:30 does not. On 5m: 08:35 contains it. `IsInWindow(T1, 5, 10)` is true from the bar stamped 08:26 to the one stamped 08:40 on 1m. On daily bars, window queries are false plus a Warning.
- **EV-8.5 Unscheduled and delayed.** 2020-03-03 10:00 FOMC: `NextEvent` at 09:59 does not return it, `PrevEvent` at 10:01 does, `IsInWindow(T1, 30, 30)` at 09:45 is false and at 10:15 true; `Classify(2020-03-03)` as-of has `hasFomc = false` and ex-post has `true`. 2020-03-15 17:00 ET: trading day = Mon 2020-03-16. NFP Sep 2025: a pre-window on 2025-10-03 08:25 is **false** (canceled at the lapse start, 2025-10-01); on 2025-11-20 08:25 the pre-window is true only after the revised date's first verifiable publication. Sep 2025 CPI: no event on 2025-10-15; 2025-10-24 08:30 present. `EventsOn(2025-10-03)` as-of omits the canceled NFP; `EventsOnExPost(…, includeCanceled: true)` returns it. Oct 2025 CPI: no event on 2025-11-13 after cancellation is known.
- **EV-8.6 Half days.** 2026-11-27 and 2026-12-24 classify `isHalfDay` with RTH close 13:15 ET; 2026-11-26 (Thanksgiving halt) has `HaltEt` 13:00; events after the close on a half day still appear in `EventsOn` but `firstReleaseEt` is unaffected.
- **EV-8.7 Across roll.** A synthetic @ES series with a splice on the CME roll Monday and another three days later: `IsRollWeek` follows the rule in both cases, `IsRollDay` follows the splice, and `NextEvent` / `MinutesSince` across the splice are unchanged (the calendar has no price state). A macro event on the splice day is reported normally. IsRollWeek uses the date-ranged historical rule (EV-1.21) for a 2018 splice.
- **EV-8.8 CSV.** Add, remove, retime (with and without a later `announced_et`), retier, `Custom:` kinds, a malformed line (fails with line number), an add row without `announced_et` fails, and a stable SHA in `Canon()`.
- **EV-8.9 Year boundaries.** `NextEvent` from 2026-12-31 16:00 ET finds the first January 2027 event. Beyond coverage it returns false plus one Warning (EV-4.7).
- **EV-8.10 Host smoke test** (in MC, **PROBE**): `_CmeSessionClockTest`-style signal `_EventCalendarTest.Strategy.CS` on @ES 1m, Exchange and Local zones, journal at Journal, a 2-year run; check the `EVENT day` lines against the CSV copy for 20 random days.
- **EV-8.11 Data probe (P-EV1).** Over 2016–2024 @ES 1m, for every T1 08:30 release, the bar with max |return| × volume in 08:25–08:40 ET is the 08:31 bar in ≥ 90% of cases, reported separately for DST, mismatch and normal weeks and for each chart zone. A failure means a time-zone or stamp error.
- **EV-8.12 Window edge and NewsBar hook.** `IsInWindow(T1, 5, 10)` is false on the 1m bar stamped 08:41 (half-open, EV-6.7). `LastT1Instant` at 08:29 returns the previous T1 instant, never 08:30; at 08:31 it returns 08:30, and at exactly 08:30:00 it returns 08:30 (at-or-before).

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9. First research uses (hypotheses only, no statistics)

These are pre-registration candidates. Nothing here is a finding (catalogue §2.6 and Gate 3 rules).

Pre-registration clause. Before any EV-9 run, register in the PR: - (a) dev window 2016-07 → 2024-12; holdout 2025+ untouched (catalogue §2.0.1); - (b) metric and test; - (c) matched control days (same weekday, same ATR tercile, no macro event, ±4 weeks); - (d) the cell list, with expected n (T1 days ≈ 32/yr → ≈ 270 dev; FOMC ≈ 68; half days ≈ 25 → descriptive only, since n < 30); - (e) Holm correction across all EV-9 cells; - (f) a kill rule; - (g) the configs charged to the trial budget.

- **EV-9.1 Event-day split for the Gate 0 canaries.** Re-read VW-08 (+ A2-24 i–iii, DV-10), A2-53, DV-01 and PS-03 outcomes split by `Classify` as-of: {T1 day, T2-only day, no macro event}, with FOMC days and half days as their own cells. *Hypothesis:* the canaries stay ≈ 0 or negative in every cell. A canary that is positive only on non-event days points to an event-day cost or fill artefact, not an edge. The split is a diagnostic. It must not be used to rescue a canary (that

would be a new config charged to the trial budget). *Metric:* net PF and mean net R per cell, block bootstrap 500/5; cells with n < 30 are report-only; a positive cell triggers a fill or cost audit, never a config.

- **EV-9.2 No-trade windows.** *Hypothesis:* a blackout of [−5, +10] min around T1 releases (Dale's rule in `trading-ideas-dale-video.md`) lowers slippage-stressed loss (A2-40/A2-59 cost model) without lowering the mean net R of level ideas. It is tested as a layer (L-step ablation) on host events. Window sizes are fixed in advance: {[−5, +10], [−15, +30], FOMC [−30, +90]}, at most 3 configs. *Metric:* p90 adverse slippage and mean net R of host events with versus without the blackout; host = the PS-01 / PS-08 / PS-24 event sets; kill if mean net R falls by more than 1 SE or slippage is not lower at Holm $p < 0.05$; 3 configs charged.
- **EV-9.3 Opening-range ideas on 08:30 days.** *Hypothesis:* on days with a T1 08:30 release the overnight range and the 09:30 IB behave differently (the release lands before RTH), so IB-based ideas should report results split by `firstReleaseEt` ∈ {08:30, 10:00, 14:00, none}. *Metric:* ON range / ATR20d and IB width / ATR20d by `firstReleaseEt` cell versus matched control days; descriptive, no trade; Holm across the 4 cells.
- **EV-9.4 Regime bucket.** *Hypothesis:* `maxTier` adds information to G-R's `AtrTercile` baseline as a stratifier. It is a prior or bucket only, never an entry trigger (same use rule as G-R, P3 §3.3). *Metric:* incremental log-loss or R^2 of `maxTier` over `AtrTercile` for next-session $|\text{return}|$ and range, dev window only; kill if the gain is ≤ 0 on the walk-forward folds.
- **EV-9.5 Roll and expiry weeks.** *Hypothesis:* level studies (A2-33) and VWAP canaries behave differently in roll weeks and on quarterly expiry and rebalance Fridays, so those sessions are reported as separate cells rather than pooled. *Cells:* roll week (rule), splice day, quarterly expiry Friday, rebalance Friday versus the rest; A2-33 edge and canary PF reported per cell; descriptive.
- **EV-9.6 A2 consumers.** A2-01/A2-05 (event AVWAP anchors CPI/NFP/FOMC release bar), A2-21 (FOMC 14:00 second leg, pooled with CPI/NFP), A2-35 (re-formation VA at T+20) and A2-20 (EIA 10:30, CL) read instants from M-EVENT post-event only (`PrevEvent` / `MinutesSince` / `LastT1Instant`).

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10. Build plan

1. `_EventCalendar.Function.CS` rules region and types; `tests/EventTests` EV-8.1, 8.3, 8.4, 8.6, 8.9 (no table yet).
2. `tools/m-event/build_table.py`; generate 2016–2027; disagreement report; EV-8.2.
3. `s_corrections` from §5.5 with sources; EV-8.5.
4. CSV loader; EV-8.8.
5. Host wiring in `_CustomSignal` (EV-6.2), journal lines, the trade-journal columns; EV-8.7 and the MC smoke test EV-8.10.
6. `_PortKeys` optional keys and publishing (EV-7).
7. Roll splice tabulation (Q3): from 2016–2024 @ES history, list every `rollOn(1)` splice date next to the date-ranged CME rule date (EV-1.21), and journal the weekday distribution and the day offset. Output `tools/m-event/out/roll_splices_2016_2024.csv`; the table goes in the PR body.
8. Data probe EV-8.11 (P-EV1) and the NewsBar hook test EV-8.12.

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11. Decisions (rev 1 open questions, resolved 2026-10-22)

Q1 tiers accepted · Q2 2016–2027, extendable to 2000 (EV-4.6) · Q3 tabulate splices from history (§10 step 7) · Q4 per EV-5.1/EV-5.3 · Q5 no entry blocking or MM action · Q6 NewsBar hook (EV-6.12), wiring in rev 1.1 · Q7 `_EconomicCalendar` untouched · Q8 journal columns only when enabled

(EV-6.10) · Q9 Jackson Hole, Chair testimony, VIX expiry, EIA WPSR added; Russell reconstitution optional (class C, off); UMich, durables, Treasury refunding not added.

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12. Changelog (rev 1 → rev 2)

12.1 Review items and how they were applied

Review item	Applied as
Blocking 1, EV-4.2 overflow	`long[] s_yyyymmddhhmm` + parallel `byte[] s_kind`
Blocking 2, EV-6.7 off-by-one	half-open `[utc − pre, utc + post)`; new edge test in EV-8.12
Blocking 3, EV-3.6 trade date	replaced with the next-session-open rule; 2020-03-15 17:00 → Mon 03-16
Blocking 4, EV-5.1 look-ahead	EV-5.1 replaced; EV-4.5 step (3b) Wayback first-publication check and step (6) raw pages under `tools/m-event/raw/`
Blocking 5, EV-5.3 shutdown	EV-5.3 replaced (canceledSince = lapse start; new date known from earliest verifiable publication); EV-5.5d/e treatment column updated
Blocking 6, Sep 2025 CPI	added to EV-5.5d, resolved from BLS notice
Blocking 7, EV-8.2 Good Friday	EV-8.2 now federal holidays only; EV-1.6 and EV-8.1 note Good Friday NFP (verified 2021-04-02, 2026-04-03)
Blocking 8, EV-2.1	`canceledSinceUtc` field and `RuleOnly` flag added (plus `Scheduled`, used by EV-4.4(f))
Blocking 9 / §9	pre-registration clause, per-item metrics EV-9.1–9.5, new EV-9.6
EV-0.4	ET instant with zone label in journal lines
EV-1.1	tentative / notation-vote rule; pre-2013 times (VERIFY)
EV-1.2	concrete per-year rule, now resolved with Fed source
EV-1.6	Good Friday sentence, dates verified
EV-1.8, EV-4.1	claims dates from DOL archive; rule only as future fallback
EV-1.12, EV-1.13	Wayback ISM history, `RuleOnly` flag
EV-1.21	date-ranged roll rule; old Thursday rule sourced; cutover VERIFY
EV-1.22	09:30 monthly SPX AM every month and 16:00 SPXW; `Quad` flag only
EV-3.3	PROBE P-EV1
EV-3.5	pre-2007 TZ note
EV-4.4	(f) `announced_et` required for add/retime
EV-4.5	steps (3b), (6), (7)
EV-5.5b	17:00 ET resolved
EV-5.5c	cancellation resolved; notice time still VERIFY
EV-5.5g	2018–19 GDP and retail rows resolved and listed
EV-5.6	extension list (VERIFY)

Review item	Applied as
EV-6.4	as-of `EventsOn` default; `EventsOnExPost`
EV-6.10	columns only when enabled; RUN line records schema
EV-7.2	`pm.evtMask` cast from `ulong`, ≤ 63 kinds asserted
EV-8.1, 8.2, 8.5, 8.7, 8.8	texts per review
New EV-8.11	08:30 spike data probe
Q1–Q9	§11; Q3 → §10 step 7; Q6 → EV-6.12 + EV-8.12; Q9 → EV-1.17/1.18/1.19/1.26/1.27

Implementation note (now specified in EV-1.24a): the new `EiaPetroleum` kind needs an `Energy` bit in `EEventClassMask` (so CL consumers can select it without ES macro windows pulling it in); Coder should add it next to `Macro` and `Calendar`.

12.2 VERIFY items remaining for the build script

1. BLS per-row announcement dates for the 2025 and 2026 lapse revisions (first Wayback capture of `2025-lapse-revised-release-dates.htm` per row); the 2026 lapse start date.
2. Sep 2025 CPI notice exact publication time (page last-modified 2025-10-10).
3. 2020-03-17/18 FOMC: whether a cancellation notice predates the 2020-03-15 17:00 statement.
4. CME roll-rule cutover date (Thursday-8-days → Monday), from Wayback captures of `rolldates.html`.
5. 2018–19 shutdown: lapse dates, BEA PCE rows, Census tip-sheet first-issue (2019-01-30) content.
6. 2025 lapse effects at BEA and Census (EV-5.5f).
7. EIA WPSR historical holiday exceptions before 2025 (Wayback), and 2016–2024 standard time.
8. Cboe VIX expiry rule and holiday handling (EV-1.26); Russell reconstitution dates (EV-1.27); Jackson Hole source page (EV-1.17).
9. Extension-only: pre-2013 14:15 / 12:30 FOMC times, 2001/2008 intermeeting actions, Oct 2013 lapse.
10. 2027 BLS/BEA schedule availability (EV-4.6).

12.3 Sources fetched 2026-10-22 for resolved items

- https://www.bls.gov/bls/2025-lapse-revised-release-dates.htm
- https://www.bls.gov/bls/092025-cpi-reschedule-notice.htm, https://www.bls.gov/news.release/archives/cpi_10242025.htm, https://www.bls.gov/cpi/additional-resources/2025-federal-government-shutdown-impact-cpi-faq.htm
- https://www.federalreserve.gov/monetarypolicy/fomchistorical2020.htm, https://www.federalreserve.gov/newsevents/pressreleases/monetary20200315a.htm
- https://www.federalreserve.gov/econres/notes/feds-notes/questions-and-answers-the-information-content-of-the-post-fomc-meeting-press-conference-20211012.html, https://www.federalreserve.gov/newsevents/2019-January.htm
- https://www.bea.gov/news/2019/initial-gross-domestic-product-4th-quarter-and-annual-2018, https://www.bea.gov/news/blog/2019-02-06/new-dates-set-gdp-personal-income-and-international-trade, https://www.census.gov/newsroom/press-releases/2019/revised-indicator-calendar.html
- https://www.cmegroup.com/trading/equity-index/rolldates.html, https://nexusfi.com/attachments/40851d1307956597, https://ohlcv.io/posts/rolling-your-own-futures-contracts/ (secondary)
- https://www.eia.gov/petroleum/supply/weekly/schedule.php

- https://www.bls.gov/news.release/archives/empsit_04022021.htm, https://www.bls.gov/news.release/archives/empsit_04032026.htm

13. Rev 2.1 rulings (agreed Analyst 1 / Analyst 2, 2026-10-22, from the PR #25 review)

- **R2.1-a (EV-4.7).** The missing-table message stays as specified: a once-per-year Warning `EVENT <year> table kinds missing: <list>`. PR B aligns the code.
- **R2.1-b (EV-3.6).** On a half day, an instant after the early halt but before 17:00 ET is out of session and maps to the next session open, the same as every other out-of-session instant.
- **R2.1-c (EV-6.7).** `EventBlackout` is Macro-only and never ORs in `Energy` implicitly. Energy consumers (for example A2-20 on CL) call `IsInWindow(..., EEventClassMask.Energy)` explicitly.
- **R2.1-d (expiry overrides)** — *superseded by the amendment below; kept for history.* ~~CME's published expiry and holiday calendar overrides the rule. Disagreements go in an exchange-calendar override table, one row per date with its CME source URL, and the build script flags every rule-vs-CME mismatch across the whole table. June 2027: the third Friday is 06-18, Juneteenth (Sat 06-19) is observed on Fri 06-18, and CME lists expiry on 06-18. The rule must not shift equity-index expiry for federal holidays that CME does not close on; fix the rule first, and use the override only if a real exception remains.~~
- **R2.1-d amended (PR27 review)** — *superseded by the 2026-10-23 amendment below; kept for history.* the premise that CME lists 2027-06-18 is false. CME's per-contract calendar (ProductCalendar/Future/133) gives ESM27 last trade 2027-06-17. The 6/18 on rolldates.html is a stale customary-date table. Equity-index quarterly expiry = third Friday, moved back to the preceding NYSE business day when NYSE is closed (SOQ requirement), per EV-1.20. The authority is the per-contract calendar, not rolldates. EV-1.20 is normative again; there is no override table.
- **R2.1-d amendment (2026-10-23).** *Amendment, approved by Ettienne Hugo 2026-10-23; reviewed and agreed by Analyst 2.* ES quarterly expiry = third Friday of the contract month; if that day is not an NYSE trading day, move back to the previous NYSE trading day. No per-contract overrides: the override table is removed and adding one is forbidden. The single NYSE holiday source is `tools/m-event/nyse_holidays.txt`, covering 2000–2035 and including historical unscheduled closures. Reference checks: ESM27 → 2027-06-17 (CME ProductCalendar); ESH08 → 2008-03-20. Roll cutover date 2022-07-01, cited to Wayback captures `20220609055445` and `20220610060819`. `cme_closed()` must be verified against CME's actual equity-futures status on historical closure days.
- **R2.1-e (EV-1.19).** EIA years outside the exception table are `RuleOnly`: no retime, the day journals `EiaRuleOnly`, and known-from is never backfilled with `MinValue`.
- **R2.1-f (EV-8.12).** The bar stamped 08:30 is still the last pre-event bar; consumers anchor on the bar that contains the instant.
- **R2.1-g (masks).** `maxTier`, `firstRelease` and `Classify` count Macro only.
- **R2.1-h (EV-4.6)** — *superseded by §14 EV-4.6b (2026-10-23); kept for history.* With `allowPartialCoverage = false`, a coverage shortfall fails: the engine is disabled and an Error line is written on the first Step (no mid-run throw).

14. EV-4.6b Per-kind, per-day fail-closed coverage

Amendment, approved by Ettienne Hugo 2026-10-23; reviewed and agreed by Analyst 2.

- **Scope.** Coverage is evaluated per event kind and per day, not per calendar.
- **Tri-state.** A day inside a coverage hole for a kind returns `Unknown` (`Event` / `NoEvent` / `Unknown`).

- **Blocking.** Unknown blocks only when that kind is in the consumer's `EEventClassMask` **and** `allowPartialCoverage = false`; otherwise it is treated as `NoEvent`.
- **Journal.** Holes are journaled once at start and once per affected day with tag `COVGAP` (via `ETraceLevel.Journal`).
- **Reporting.** Report, per kind and per test window, the share of days blocked due to `Unknown`; flag any window above 5%.
- **Engine-wide disable.** Applies only when the table file is missing or its hash mismatches. This replaces the previous whole-calendar-off default for gaps (R2.1-h).
- **Required tests.** (1) a hole day returns `Unknown`; (2) it blocks under `allowPartialCoverage = false`; (3) it is treated as no-event under `true`; (4) end-of-coverage (e.g. JOLTS after 2026-12) behaves the same as a hole; (5) a hole day for a kind NOT in the consumer's `EEventClassMask` does not block, even under `allowPartialCoverage = false`.
- **Tier filter (added 2026-10-23, reviewed by Analyst 2).** When a query passes a minimum tier, an `Unknown` kind whose effective tier is below that minimum does not block, since it could never have opened the window. The effective tier is resolved from the same config the query runs under (`tierOverride` applied first, then compared), not from the table default. Such a day still gets its `COVGAP` journal entry but does not count toward the >5% Unknown-blocked share. Required test (6): an `Unknown` hole for a kind below the query's minimum tier does not block under `allowPartialCoverage = false`, and the same hole at or above the minimum does.
- **Follow-up data task (out of scope for this build).** Fill the 80 GDP/PCE 2016–2024 holes from BEA release archives.